

ESP32-S3 Train GPS Tracker

Pin Connection Documentation & Wiring Schematic

MCU:	ESP32-S3 DevKit (38-pin)
Display:	ILI9488 3.5" TFT 480x320 with XPT2046 touch
Audio:	MAX98357A I2S mono amplifier
GPS:	L89H (UART, 9600 baud)
Storage:	MicroSD card (SPI mode)

1. Complete GPIO Assignment

GPIO	Signal	Connected To	Dir	Notes
1	BAT_ADC	Voltage divider mid-point	IN	100kOhm/100kOhm from VBAT -- ADC 11dB
5	I2S_BCLK	MAX98357A - BCLK	OUT	I2S bit clock
6	I2S_LRC	MAX98357A - LRC/WS	OUT	I2S word select
7	I2S_DIN	MAX98357A - DIN	OUT	I2S serial audio data
8	TFT_DC	ILI9488 - DC/RS	OUT	Data/Command select
9	TFT_RST	ILI9488 - RESET	OUT	Active LOW reset
10	TFT_CS	ILI9488 - CS	OUT	TFT chip select
11	SPI_MOSI	TFT + Touch + SD (shared)	OUT	Shared SPI data out
12	SPI_SCK	TFT + Touch + SD (shared)	OUT	Shared SPI clock
13	TOUCH_MISO	XPT2046 - MISO (DO)	IN	Touch only -- TFT SDO NOT connected
14	TOUCH_CS	XPT2046 - CS	OUT	Touch chip select
15	SD_CS	SD card module - CS	OUT	SD chip select
16	GPS_RX	GPS module - TX	IN	UART1 RX
17	GPS_TX	GPS module - RX	OUT	UART1 TX
40	SD_MISO	SD card module - MISO	IN	Separate MISO for SD

NOTE: ILI9488 SDO pin must NOT be connected -- it does not tristate when CS is high and would conflict with XPT2046 MISO on GPIO13.

2. Per-Peripheral Wiring

2a. TFT Display -- ILI9488 3.5" (480x320)

Display Pin	Label	ESP32-S3 GPIO	Notes
VCC	3.3V	3V3	
GND	GND	GND	
CS	LCD_CS	GPIO 10	
RESET	LCD_RST	GPIO 9	Active LOW
DC/RS	LCD_DC	GPIO 8	Data/Command select
SDI (MOSI)	LCD_MOSI	GPIO 11	Shared SPI bus
SCK	LCD_SCK	GPIO 12	Shared SPI bus
LED/BL	Backlight	3V3 or via 330Ohm	Always-on backlight
SDO (MISO)	--	Do NOT connect	Bus conflict risk

2b. Touch Controller -- XPT2046 (on TFT board)

Touch Pin	Label	ESP32-S3 GPIO	Notes
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T_CLK	SCK	GPIO 12	Shared SPI bus
T_CS	CS	GPIO 14	Own chip select
T_DIN	MOSI	GPIO 11	Shared SPI bus
T_DO	MISO	GPIO 13	Separate MISO line
T_IRQ	IRQ	--	Not used (polling mode)

2c. SD Card Module

SD Pin	Label	ESP32-S3 GPIO	Notes
VCC	3.3V	3V3	
GND	GND	GND	
CS	CS	GPIO 15	Own chip select
MOSI	MOSI	GPIO 11	Shared SPI bus
CLK	SCK	GPIO 12	Shared SPI bus
MISO	MISO	GPIO 40	Separate MISO line

2d. Audio Amplifier -- MAX98357A (I2S Mono)

MAX98357A Pin	Label	ESP32-S3 GPIO	Notes
VIN	5V	5V (VUSB/boost)	Do NOT use 3.3V
GND	GND	GND	
BCLK	BCLK	GPIO 5	I2S bit clock
LRC / WS	LRC	GPIO 6	Word select
DIN	DIN	GPIO 7	Serial audio data
SD (shutdown)	--	Float or 3.3V	Float = enabled
GAIN	--	Float	9 dB gain (default)
OUT+ / OUT-	SPK	4-8 Ohm speaker	Direct speaker output

2e. GPS Module -- L89H (UART)

GPS Pin	Label	ESP32-S3 GPIO	Notes
VCC	3.3V	3V3	Check module datasheet -- some need 5V
GND	GND	GND	
TX	GPS TX	GPIO 16	GPS TX -> ESP32 RX
RX	GPS RX	GPIO 17	GPS RX <- ESP32 TX
PPS	--	--	Not used

NOTE: UART1, 9600 baud, 512-byte RX buffer. GPS TX must connect to ESP32 GPIO16 (RX), not TX.

2f. Battery Monitor (Voltage Divider on GPIO 1)

ADC Raw	GPIO 1 Voltage	Battery Voltage	State
~2610	~2.10 V	4.2 V	Fully charged
~2298	~1.85 V	3.7 V	Nominal
~1862	~1.50 V	3.0 V	Empty (protect cell)

NOTE: Voltage divider: VBAT -> R1(100 kOhm) -> GPIO1 -> R2(100 kOhm) -> GND. Call analogSetAttenuation(ADC_11db) in setup().

3. Shared SPI Bus & MISO Switching

The TFT display, XPT2046 touch controller, and SD card all share one SPI bus (SCK=GPIO12, MOSI=GPIO11). However each device has its own MISO line, so the firmware switches the active MISO in software before each access using SPI.end() followed by SPI.begin() with the correct MISO pin. Each device also has its own dedicated CS pin to ensure only one is active at a time.

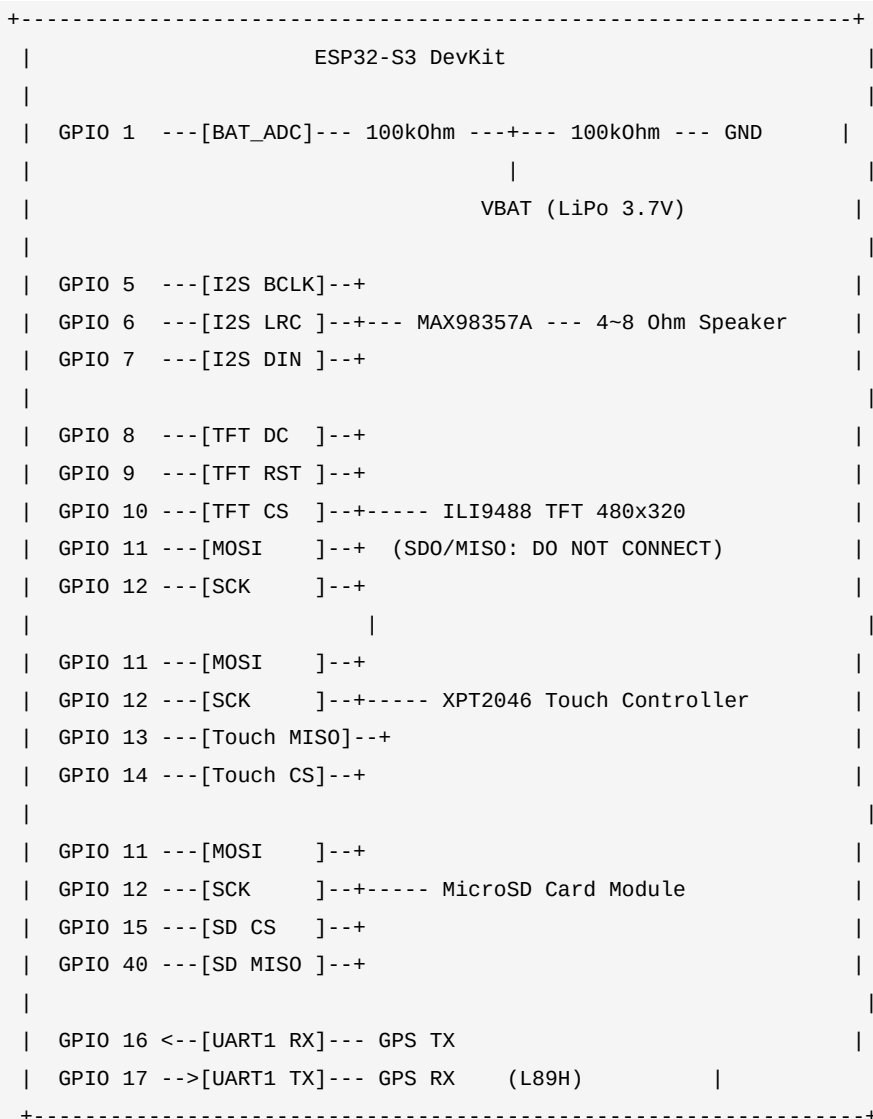
Before SD access:

```
SPI.end();
SPI.begin(12, GPIO40, 11); // SD_MISO = GPIO40
```

Before Touch access:

```
SPI.end();
SPI.begin(12, GPIO13, 11); // TOUCH_MISO = GPIO13
```

4. Block Schematic Diagram



SPI Bus Summary:

SCK (GPIO 12) -----> TFT, Touch, SD (all 3 share this)

MOSI (GPIO 11) -----> TFT, Touch, SD (all 3 share this)

MISO switched by FW:

GPIO 13 <----- XPT2046 MISO (active during touch reads)

GPIO 40 <----- SD MISO (active during SD access)

TFT SD0 = NOT CONNECTED

5. Power Supply Summary

Module	Voltage	Typical Current	Source
ESP32-S3	3.3 V	240 mA (WiFi on)	DevKit on-board LDO
ILI9488 TFT	3.3 V	30-80 mA	ESP32 3V3 pin
XPT2046	3.3 V	~2 mA	ESP32 3V3 pin
SD Card	3.3 V	100-200 mA active	ESP32 3V3 pin
GPS L89H	3.3 V	~50 mA	ESP32 3V3 pin
MAX98357A	5 V	Up to 500 mA peak	USB or 5V boost
LiPo Battery	3.7-4.2 V	-- (source)	External charger

NOTE: The ESP32-S3 DevKit LDO can typically supply ~500 mA total from the 3V3 pin. If all peripherals are active simultaneously, consider an external 3.3V regulator.

NOTE: MAX98357A must be powered from 5V. Use a 5V boost converter if running from LiPo battery.

6. TFT_eSPI User_Setup.h (key settings)

```
#define ILI9488_DRIVER

#define TFT_MISO 13    // XPT2046 MISO -- TFT SDO NOT connected
#define TFT_MOSI 11
#define TFT_SCLK 12
#define TFT_CS  10
#define TFT_DC   8
#define TFT_RST  9

#define TOUCH_CS 14    // XPT2046 chip select
```